Brighter Blue

Core Testing>Accelerators & ATA Basics

Student Guide

IBM

Brighter Blue

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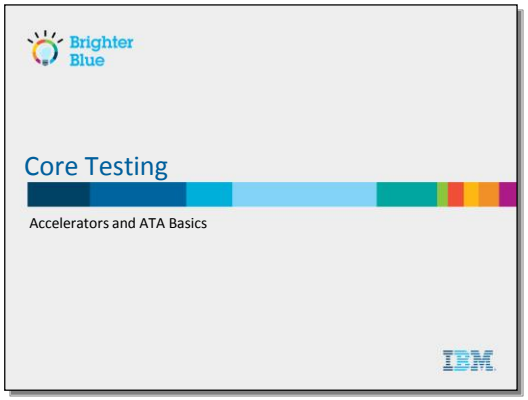
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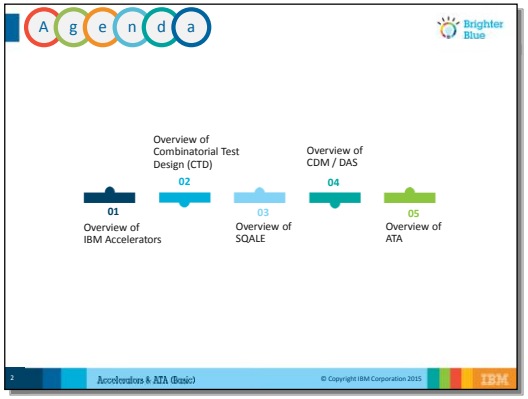
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Module 01: IBM Accelerators Overview

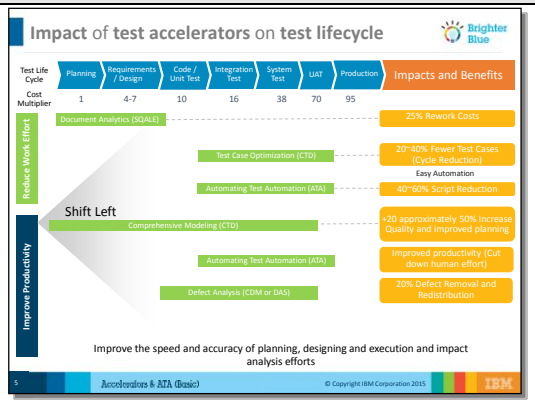
Slide Content	Use this column for your notes
Slide 1 	

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Slide 2  **At the end of this course, you should be able to:** ▪ Describe IBM Accelerators, such as CTD, SQALE, CDM, and ATA ▪ Identify the use of test accelerators on test life cycle. ▪ Define the basic concept of Modeling, specially CTD ▪ Define the basic concept of Document Analytics ▪ Identify the benefits and barriers of test automation ▪ Describe the usage of ATA	

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Slide 3  At the end of this module, you should be able to: ▪ Recognize the features of IBM Total Test Quality (TTQ) approach ▪ Differentiate between the Traditional test approach and next generation testing	
Slide 4	

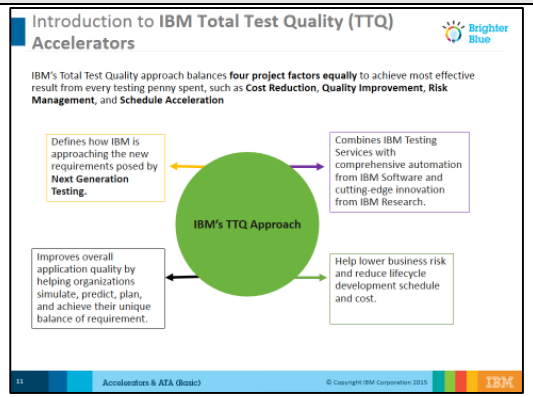
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Core Testing>Accelerators & ATA Basics

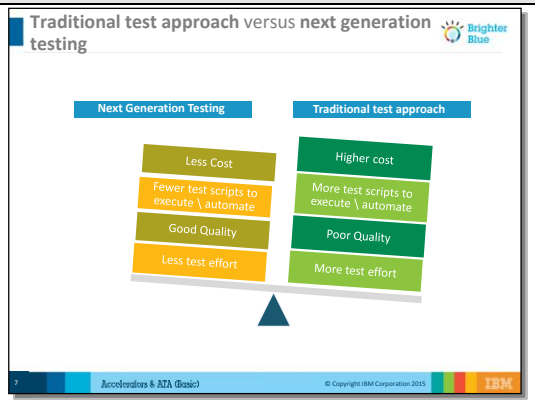
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 **Note** that this diagram is a depiction of high level requirements and the design figures are derived from Boehm and Capers-Jones. The Detailed Requirements Review and Detailed Design Review multipliers are from IBM Research and were developed subsequent to when the earlier models were published, whereas savings are based on solution benchmarks. Earlier, we had either manual or automated testing but, now there are different levers of testing in the software development lifecycle. An appropriate testing shifts the defect finding activities to the left of SDLC thus reducing total cost of development dramatically.	

Core Testing>Accelerators & ATA Basics

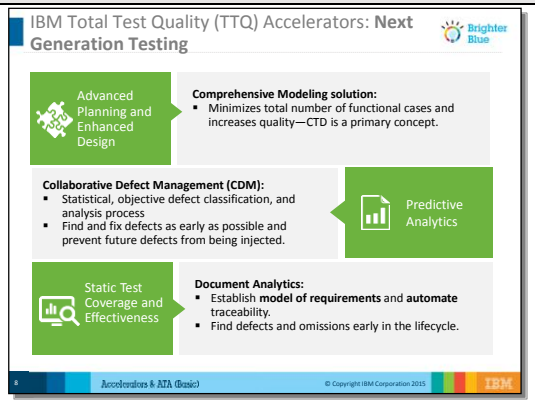
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It costs more to fix a defect that is identified at a later stage of project lifecycle than the one which is identified early. However, early testing offers several challenges. IBM understands these challenges and offers several accelerators that are aligned to different phases of testing. These accelerators ensure that we test early (shift left) and test less that prevents critical defects. Some **examples of testing accelerators** are: ▪ An accelerator that minimizes the number of test cases. ▪ Adaptive Test Automation **(ATA) –** a test automation accelerator that automatically converts manual test steps in executable steps in natural language.	
Slide 6	

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 IBM's Total Test Quality approach balances **four project factors equally** to achieve most effective result from every testing penny spent, such as **Cost Reduction**, **Quality Improvement**, **Risk Management**, and **Schedule Acceleration**. **IBM's TTQ approach:** ▪ Defines how IBM is approaching the new requirements posed by **Next Generation Testing.** ▪ Combines IBM Testing Services with comprehensive automation from IBM Software and cutting-edge innovation from IBM Research.	

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▪ Help lower business risk and reduce lifecycle development schedule and cost. ▪ Improves overall application quality by helping organizations simulate, predict, plan, and achieve their unique balance of requirement.	
Slide 7	

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 In a traditional approach, there are more number of test cases, add more testers, and more test scripts needs to be automated. This takes more time, effort, adds to cost and has a low ROI. The next generation testing or intelligent testing approach—reduces number of test cases and reduce number of testers. This results faster go to market, less cost, and high ROI. In addition, this type of Testing has fewer number of test cases to be automated.	
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Core Testing>Accelerators & ATA Basics

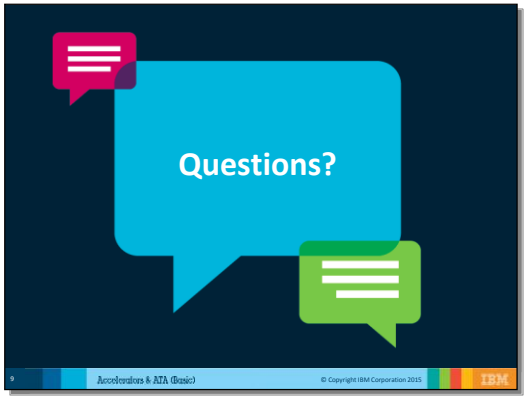
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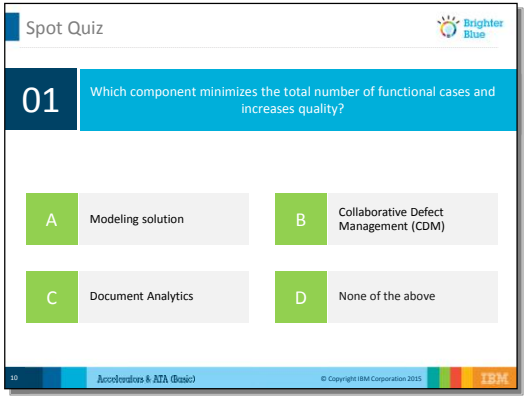
Next generation testing has been made possible through the collaboration with IBM research and use of innovative test accelerators, such as and Adaptive Test Automation (ATA).

CTD is an accelerator that minimizes the number of test cases and **ATA** is a test automation accelerator that automatically converts manual test steps to executable steps in natural language.

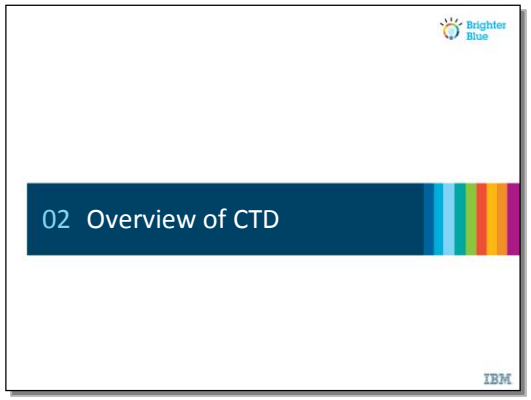
Software development and testing organization need assistance as they are increasingly involved in critical projects with tight deadlines and low budget. And, the situation goes worse when there is a very large project. So, sometimes, research has seen that the cost of testing takes more than 60 or

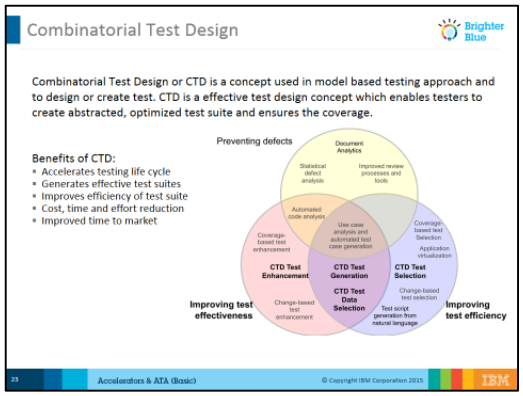
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70 % time of entire SDLC. Even after that, people feel that the problem is due to defect leakages. Besides that, current testing activities are ad-hoc and it depends upon the personal experience to explore systems, time, and cost. **Features of TTQ** **Comprehensive modelling solution:** Unlike development, system testing is not model or module-based or components-based activities, rather it is flat. So, when we are moving towards next generation testing from traditional-based approach, exhaustive testing in the real world is not enough. Besides, it takes billion and trillion of test cases. Hence, we are trying to bring more modularity-module and model based testing. These are more components driven, function point and test-point driven in terms of models. Modeling solution helps minimize total number of functional cases and can increase quality of the **Collaborative Defect Management (CDM):** This helps in the statistical, objective defect classification, and analysis process. It also helps to find and fix defects as early as possible and prevent future defects from being injected. **Document Analytics**: This helps to establish model of requirements, automate traceability and to find defects and omissions early in the lifecycle, to	

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minimize the cost to correct, the impact on schedule, and the amount of rework.	
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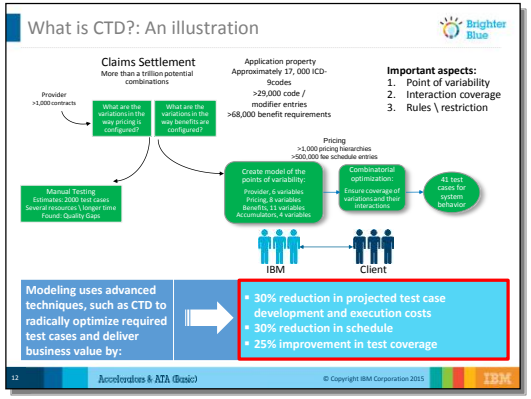
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Module 02: Overview of Combinatorial Test Design (CTD)

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Slide 11  At the end of this module, you should be able to: ▪ Define CTD ▪ Explain the common flow for modeling ▪ List the different types of models	

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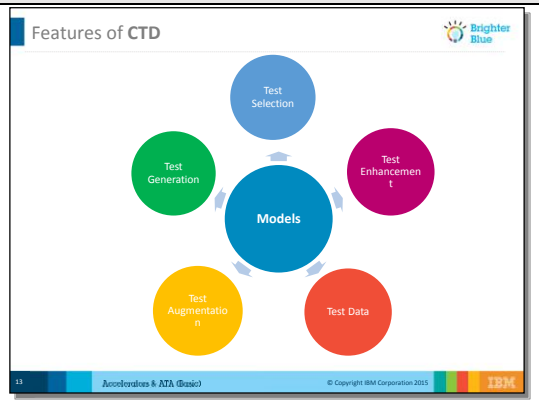
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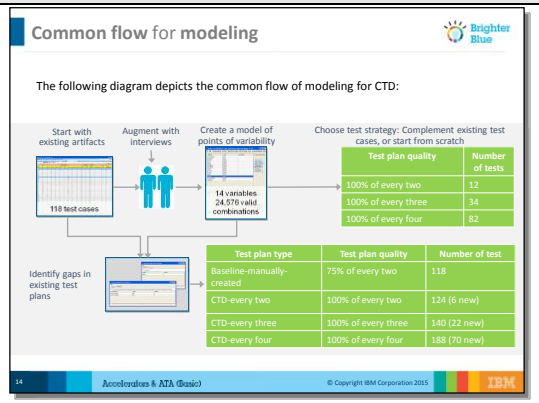


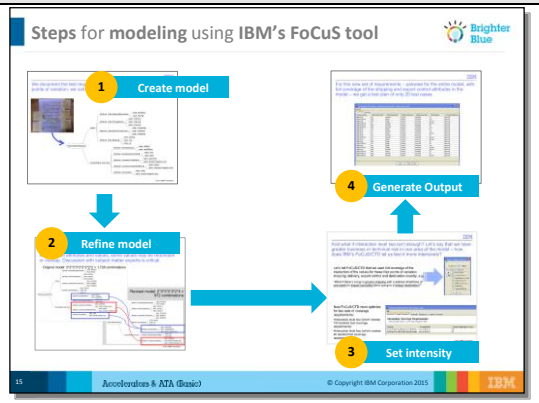
CTD is a primary concept of modeling solution in the next generation accelerators. This slide provides an illustration to explain the concept of CTD. Some of the important aspects of CTD or modeling are:

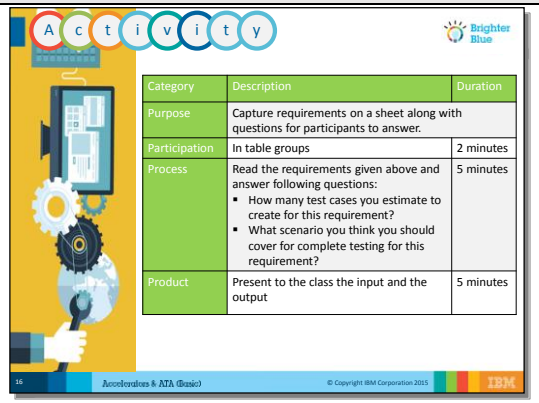
•**Point of variability:** When we get a requirement initially from the clients or business source, such as Claims Settlement so one can deduce if someone would like to create test cases and come up with millions and trillions of potential combinations because in claims settlement, there are thousands on

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contracts for a policy. Similarly, there are different combinations of application properties. For example, there can be 68,000 benefit requirements, and so on. Consider a scenario of manual testing in a large account, the existing tester has created around manual test cases as they initially estimated for 2000 test cases with the demand for several resources for 5-6 months and there was a quality gap as well. •Interaction coverage •Rules \ restriction CTD is an innovative approach, which yield greater business benefits. It helps reduces risk through increased coverage and improves quality. It also minimizes the number of test cases, which helps in reduction of cost, time, and resources. Based on the research after doing lots of work on several applications of similar type in large engagements, it has been estimated that CTD helps reduce the **projected test case development** and **execution costs** by 30%. This helps in reduction of 30% in the schedule, and also we have come up with 25% of increment in the test coverage.	
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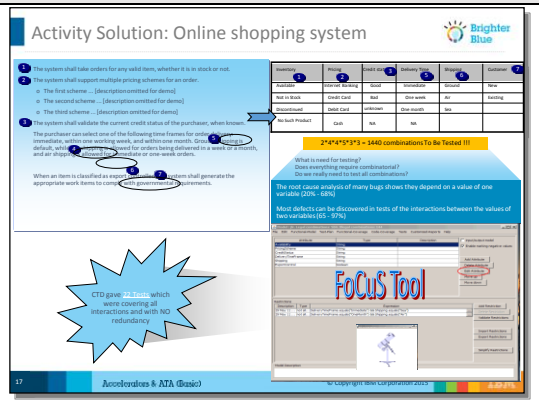
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 There are different types of models used in CTD. Let us have a hi ▪ Create or identify a lean subset of test to cover a similar functionality. ▪ Create a lean solution recommendation to gain higher quality with less test. ▪ Used for maintenance purpose and enhance existing model to alter or cover more. ▪ Add more to existing model to cover gap. **Test Data Model:** Create Test Data Model for setting up optimized data.	
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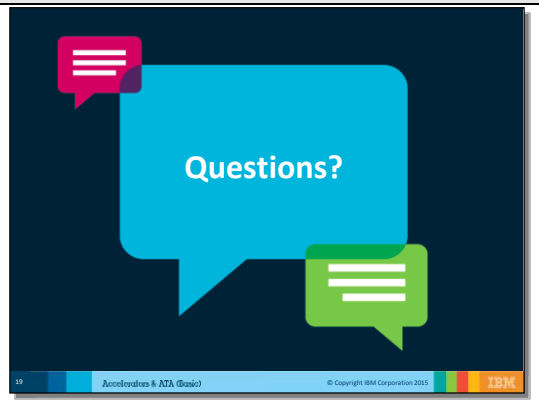
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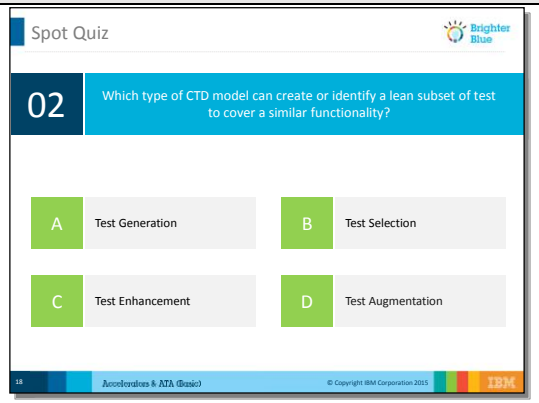
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 The four steps of modeling are: 1. Create model 2. Refine model 3. Set intensity 4. Generate plan	
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Slide 18	

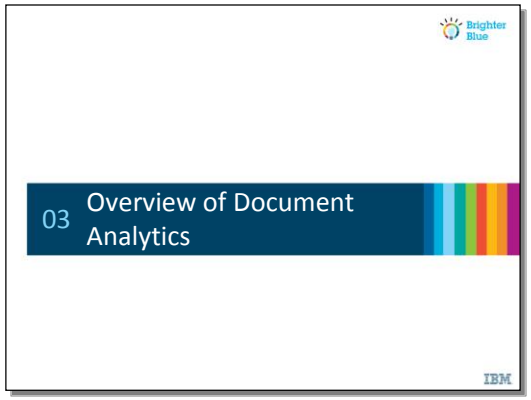
Core Testing>Accelerators & ATA Basics

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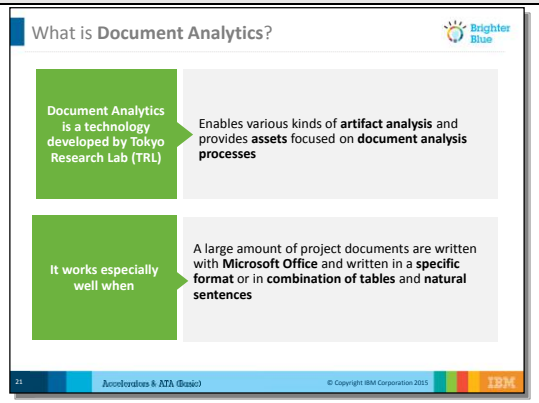
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Module 03: Overview of Document Analytics

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Slide 21  At the end of this module, you should be able to: ▪ Recognize the features of the Document Analytics tool ▪ List the advantages of the Document Analytics tool ▪ Identify the steps performed by the Document Analytics tool to analyze documents ▪ Identify the usage and examples of Document Analytics	

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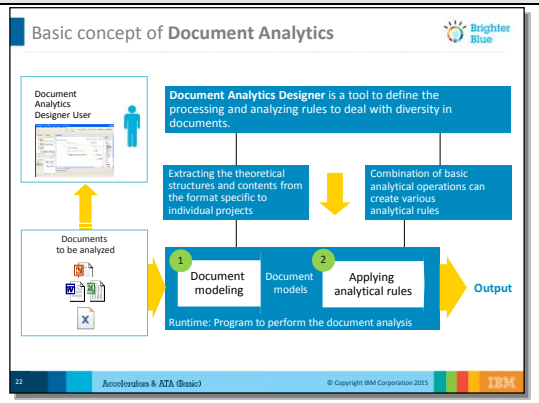
Document Analytics technology is a solution, developed by to enable analysis of diverse project artifacts (documents) applying advanced document modeling and text analysis.

In particular, it provides assets focused on document analysis processes that lack analytical technologies so that various kinds of documents (requests, design and test documents during early phases of the project life cycle) can be effectively processed.

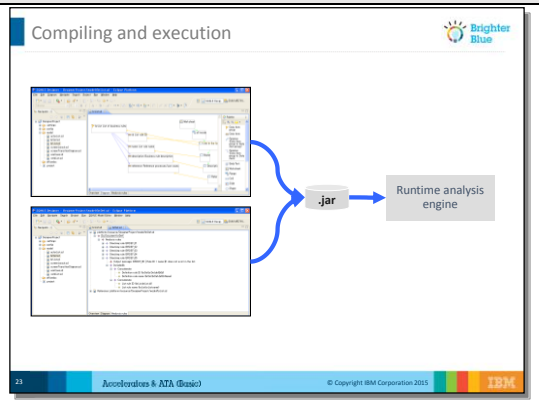
It works especially well when the following situations exist:

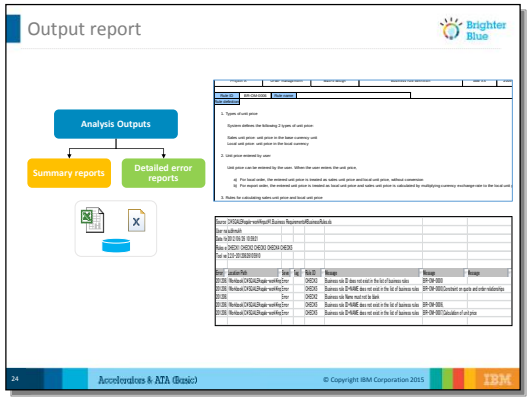
- A large amount of project documents are written with Microsoft Office™.

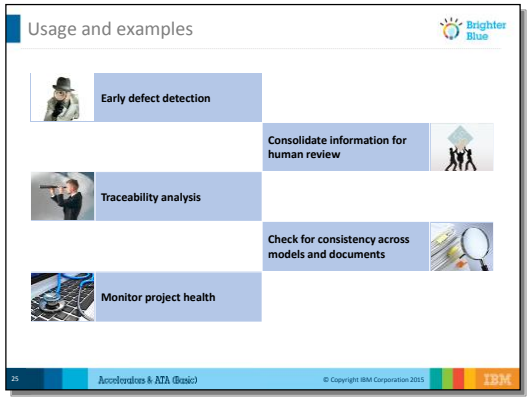
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▪ Documents are written in a specific format. ▪ Documents are written in combination of various tables and natural sentences. ▪ When analysis must be performed over multiple documents and/or various kinds of deliverables. Artifacts: All constructs created during the life cycle process of a project, including documents and code. These constructs include not only deliverables, such as products delivered to customers, but also intermediates and information necessary for process control.	
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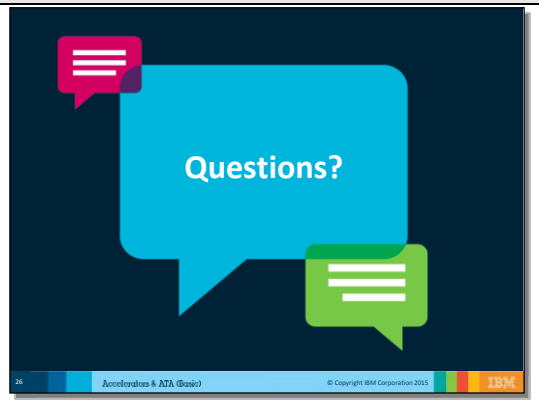
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 Document Analytics helps detect simple but time-consuming defects if handled manually, and allows you to concentrate on the more complicated defects. **Document Analytics Designer** is an Eclipse workbench that provides a visual modeling tool to build document models as well as creating analytical rules. The user who works on the **Document Analytics Designer** should define the rules on data extractions, analyses and conversions on GUI depending on individual requests. Note that you do not need any programming for Document Analytics Designer.	

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Document Analytics performs document analysis using the following two steps: ▪ **Document modelling**: Document Analytics extracts the content, by mapping the structure and content of each different type of document that is in scope. The result of this is a Document model. ▪ **Application of analytical rules**: Document Analytics applies analysis rules to the extracted document models to modify or verify the models.	
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 Once the document modeling is complete and corresponding analytical rules are applied, we now run the Document Analytics using compiling and execution. This creates a Java code that is deployed to the Runtime analysis engine also known as Document Analytics runtime. Then, Document Analytics can be run on the a variety of artifacts on Document Analytics runtime by schedule and / or on demand, which then creates outputs like summary reports, detailed error reports and newly generated artifacts.	

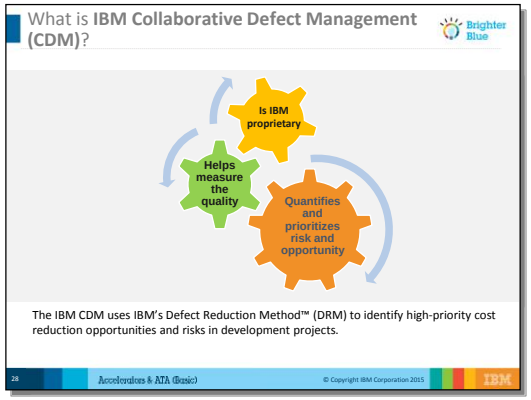
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Slide 25  Document Analytics can be run on a variety of artifacts on an ongoing basis. This tool generates different types of output reports, which include: • Summary reports: These types of reports contain the various information, such as number of document files, size metrics of each file, and number of defects. • Detailed error reports: These types of reports contain the following information: severity, type of issue (rule ID), problem description, and links to the error locations. New artifacts generated.	

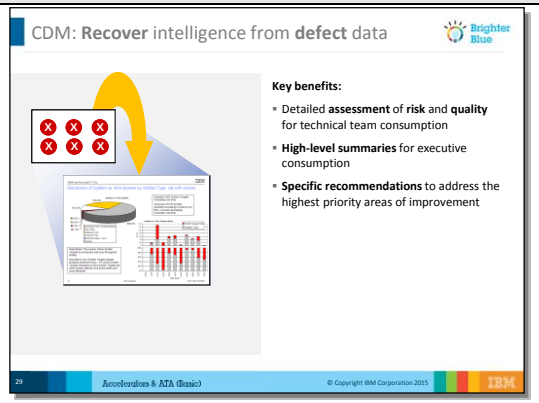
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Slide 26  Some of the usage and examples of Document Analytics include: **Early defect detection:** Finding errors in requirements and design documents or during test plan or test case creation phase. **:** Creating new documents using information from multiple sources : Checking relationships across development and test : Comparing business process models with specification and design documents : Measuring document quality over time	
Slide 27	

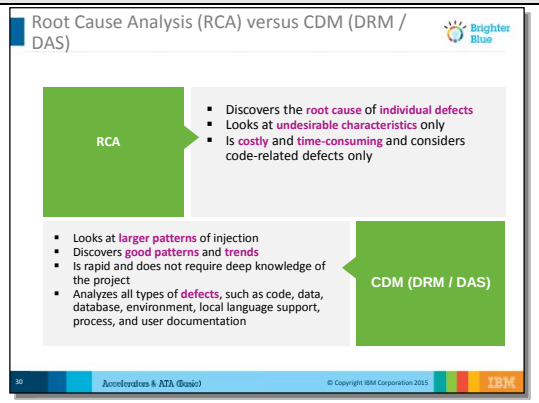
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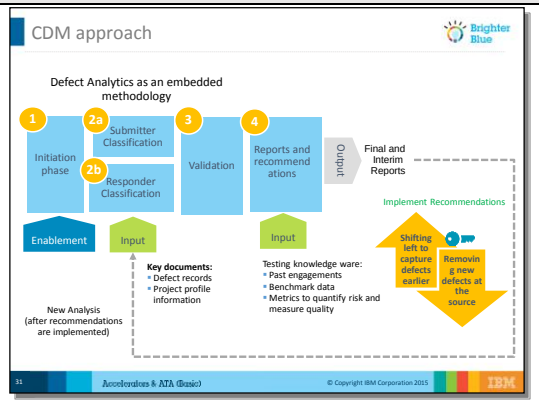
Module 04: Overview of CDM / DAS

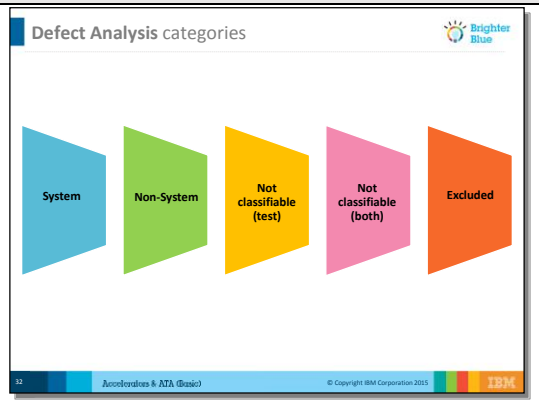
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Slide 28  At the end of this module, you should be able to: ▪ Describe the features of the CDM tool ▪ Identify the benefits of using the CDM tool ▪ Differentiate between Root Cause Analysis (RCA) and CDM (DRM / DAS) ▪ Identify the phases involved in the CDM approach ▪ Identify the different categories of defect analysis ▪ List the milestones involved in the phases of CDM	

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Slide 29  CDM uses IBM's Defect Reduction Method™ (DRM) to identify high-priority cost reduction opportunities and risks in development projects. CDM: ▪ IBM proprietary ▪ Based on over two decades of IBM research ▪ Objectively quantifies and prioritizes risk and opportunity ▪ Objectively measures quality	
Slide 30	

Slide Content	Use this space for your own notes
 The IBM Collaborative Defect Management provides business value: ▪ Recommend action items that reduce risk and cost associated with development and testing ▪ Acknowledge and reinforce client strengths ▪ Introduce clients to Testing Services offerings that increase ROI and improve quality	
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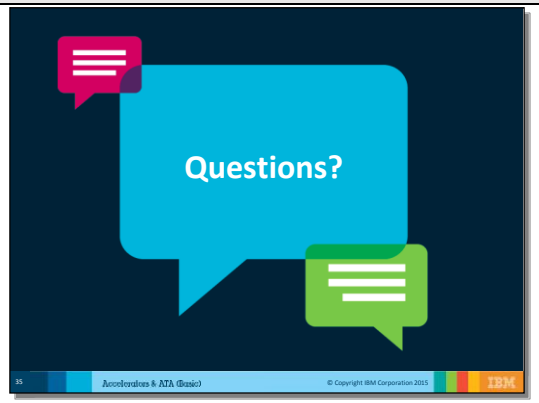
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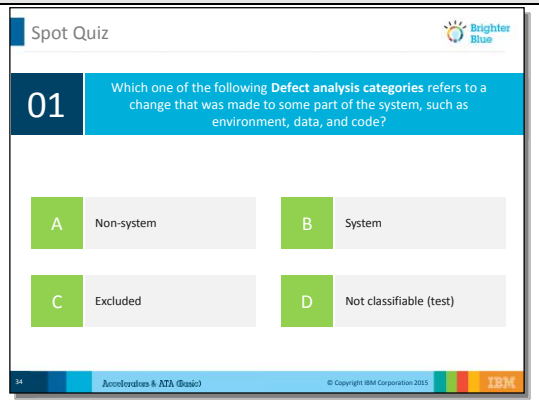
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 Defect Analytics is used as an embedded methodology	
Slide 33	

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 Some of the Defect Analysis categories are listed below: ▪ **System**: This category refers to a change was made to some part of the system, such as environment, data, code, and so on as a result of the defect ▪ **Non-System**: This category refers to a change was made to the way test was performed, either implicitly or explicitly, as a result of the defect ▪ **Not classifiable (test)**: Insufficient information was provided to classify the type of test that uncovered the defect ▪ **Not classifiable (resolution)**: Insufficient information was provided to classify the fix that was implemented	

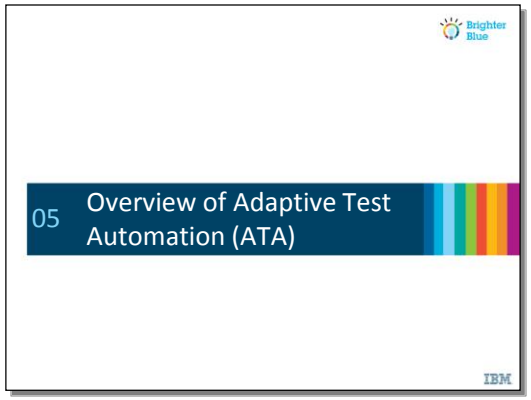
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▪ **Not classifiable (both)**: Insufficient information was provided to classify the type of test that uncovered the defect and the fix that was implemented ▪ **Excluded**: The defect was not in a final (Closed) state, a final resolution was either not yet in place or not yet verified, or defect was out of scope Non-System Artifact Types: ▪ Verification Point / Expected Results ▪ Data Parameters / Inputs ▪ Test Steps ▪ Test Case ▪ Execution Plan ▪ Resolution Plan	
Slide 34	

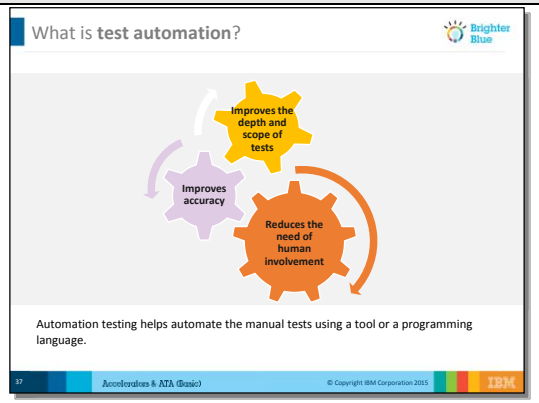
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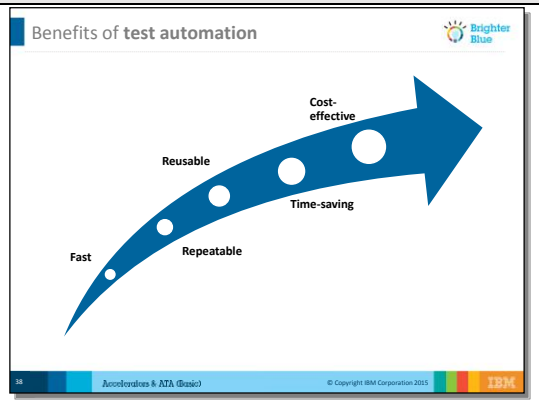
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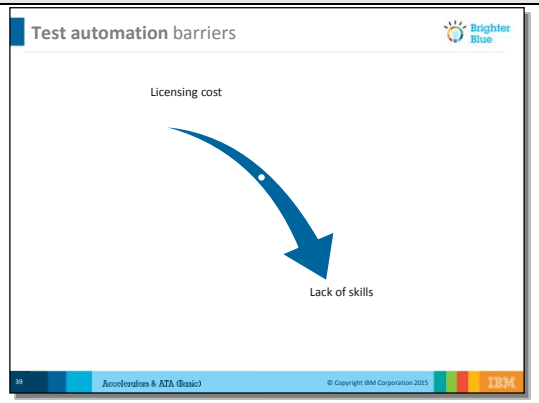
Module 05: Overview of Adaptive Test Automation (ATA)

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Slide 37  At the end of this module, you should be able to: ▪ Identify the benefits and barriers of test automation ▪ Recognize the features of ATA ▪ Identify the different variants of ATA ▪ Identify the features of the high-level architecture of ATA ▪ List the assumptions of implementing ATA	
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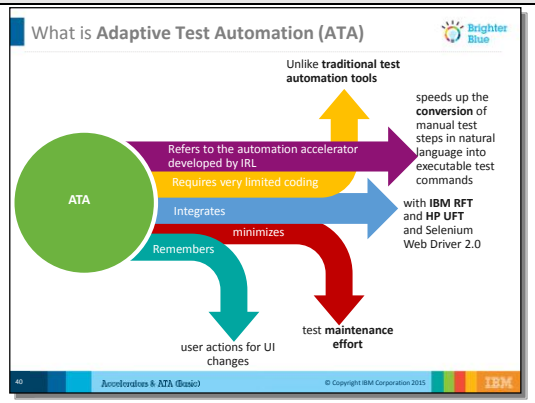
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 The benefits of test automation are listed here: **Fast**: The automated tests run fast as compared to the tests run by the humans as these kinds of tests are run by the tools. **Repeatable**: Test automation allow you to re-run the tests in a similar manner, which eliminates the risk of any human error. **Reusable**: You can reuse automated tests on different versions of the software irrespective of a change in the user interface. **Time-saving**: Test automation allows you to save time by reducing the time taken to run tests especially in case of regression testing and enable you to focus on tasks that are difficult and cannot be automated. **Cost-effective**: Automation testing also helps reduces the number of resources required for regression testing.	

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Source: http://red-badger.com/blog/2013/02/01/benefits-of-automated-testing/	
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 The key barriers of test automation are listed here: **Licensing cost:** One of the biggest barrier of automation testing is the high cost of licensing associated with the automation tool. **Lack of skills:** Automation tools are script-based, and require programming expertise. Usually, organizations do not have employees with the required skills and need to be trained or hire new employees that adds to the cost.	
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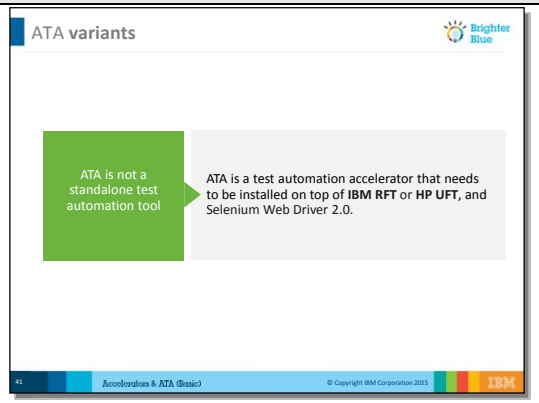


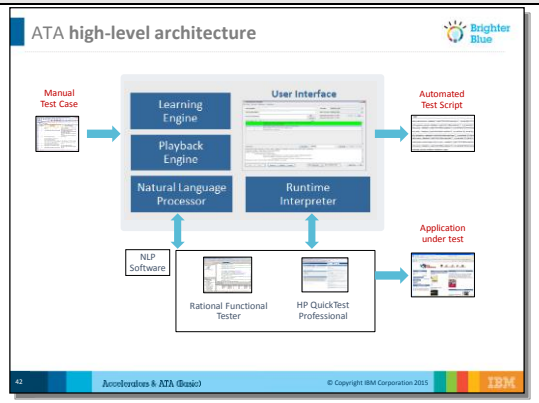
It is a test automation accelerator developed by IRL (IBM India Research Lab) that speeds up the conversion of manual test steps in natural language into executable test commands.

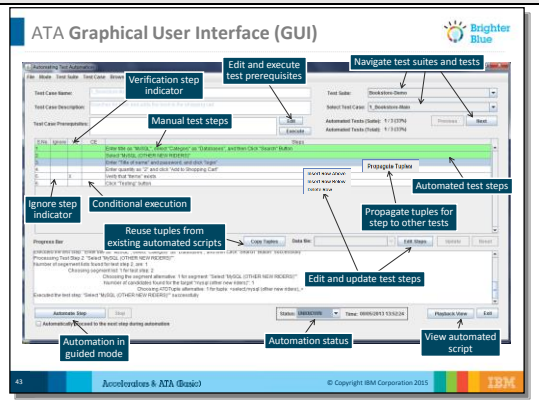
Unlike traditional test automation tools, ATA requires very limited coding (coding is needed only for test steps that involve custom verification logic).

ATA is fully descriptive and does not store any test object. This minimizes test maintenance effort. ATA offers a unique feature called “Proximity-based Label Association” which ensures that during test execution ATA is able to find UI objects even after these have changed, by referring other UI objects which are associated with it. This further reduces maintenance effort.

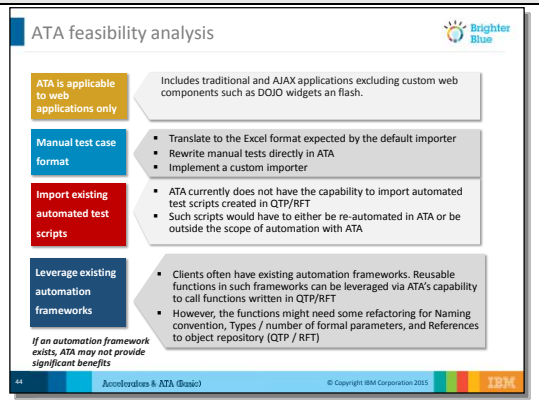
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ATA learns from “user feedback” that is it remembers user actions for UI changes and remembers these to avoid further user actions for similar changes. ATA integrates with two industry standard tools—IBM RFT and HP UFT. ATA is easy to learn, master and implement and saves considerable test automation effort. ATA has already been proposed for some of the large test automation deals at IBM.	
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 ATA is not a standalone test automation tool such as IBM RFT or HP UFT. It is a test automation accelerator that needs to be installed **on top** of one of these tools, and requires Selenium Web Driver 2.0 to be installed on the machine. This means that one of these tools should already been in place (installed and licensed) before ATA can be installed.	
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ATA is applicable to web applications only, which includes traditional and AJAX applications excluding custom web components such as DOJO widgets an flash):

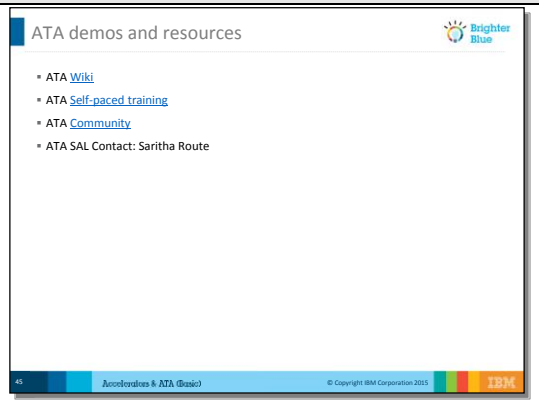
- ATA-QTP: Any web application that can be automated using the web add-in in QTP can also be automated using ATA-QTP
- ATA-RFT: Any web application that can be automated without extra plug-ins in RFT can also be automated using ATA-RFT

Manual test case format:

- Translate to the Excel format expected by the default importer
- Rewrite manual tests directly in ATA
- Implement a custom importer

Importing existing automated test scripts:

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▪ ATA currently does not have the capability to import automated test scripts created in QTP/RFT ▪ Such scripts would have to either be re-automated in ATA or be outside the scope of automation with ATA Leveraging existing automation frameworks: ▪ Clients often have existing automation frameworks ▪ Reusable functions in such frameworks can be leveraged via ATA's capability to call functions written in QTP/RFT ▪ However, the functions might need some refactoring: ○ Naming convention ○ Types/number of formal parameters ○ References to object repository (QTP/RFT) If an automation framework exists, ATA may not provide significant benefits	
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